Case Study: Norfolk, Virginia - Integrated Coastal Resilience Planning

Description	uses multiple tools integerated approach
Related Urban Climate Tools	Climate Central Coastal Risk Screening Tool, Sea Level Rise Viewer (NOAA)

Norfolk's downtown waterfront facing chronic tidal flooding - photo showing flooded street during king tide

Executive Summary

Location: Norfolk, Virginia, USA

Population: 238,000 (metro: 1.8 million)

Climate Challenge: Recurrent tidal flooding, sea level rise, storm surge vulnerability

Project Duration: 2014 - ongoing

Total Investment: $2.7 billion (approved through 2040)

Tools Used: Multiple climate adaptation platforms for assessment, planning, and community engagement

Norfolk, a historic naval city on the Chesapeake Bay, faces some of the highest rates of relative sea level rise on the U.S. East Coast (4.59mm/year - nearly twice the global average). The city experiences regular tidal flooding even on sunny days, and Hurricane Irene (2011) and Sandy (2012) caused severe damage. This case study examines how Norfolk used publicly accessible climate adaptation tools to transform from reactive emergency response to proactive, integrated resilience planning.

Background & Context

The Challenge

Norfolk's vulnerability stems from multiple compounding factors:

- **Rapid sea level rise**: 1.5 feet since 1930; projected 1.5-2 feet more by 2050
- **Subsidence**: Land is sinking due to groundwater extraction and post-glacial adjustment
- **Low elevation**: Much of the city is less than 10 feet above mean sea level
- **Dense development**: Limited space for traditional gray infrastructure solutions
- **Economic constraints**: Medium-income city with limited tax base
- **Military importance**: Home to Naval Station Norfolk (world's largest naval base)
- **Historic neighborhoods**: Preservation concerns limit adaptation options
- **Environmental justice**: Low-income and minority neighborhoods face disproportionate flood risk

By 2014, Norfolk was experiencing tidal flooding 10+ days per year in downtown areas. Residents and businesses were growing frustrated with chronic "nuisance flooding" disrupting daily life. The city needed a comprehensive adaptation strategy backed by solid data.

Stakeholders Involved

- **City of Norfolk**: Planning, Public Works, Emergency Management departments
- **Old Dominion University**: Coastal research and modeling support
- **U.S. Navy**: Naval Station Norfolk facility planning
- **Community Organizations**: Neighborhood associations, environmental justice groups
- **Private Sector**: Real estate developers, insurance companies, utilities
- **Regional Partners**: Hampton Roads Planning District Commission
- **Federal Agencies**: FEMA, NOAA, U.S. Army Corps of Engineers

- **Foundations**: Kresge Foundation, Rockefeller Foundation (100 Resilient Cities)

Tools & Methodologies Used

Assessment & Data Tools

1. NOAA Sea Level Rise Viewer

- **Application**: Initial citywide vulnerability screening
- **How it was used**: Planners mapped which neighborhoods would face inundation under different sea level rise scenarios (1-6 feet). Created baseline maps showing critical infrastructure at risk.
- **Outputs**: Identified priority areas for detailed study; justified need for resilience planning to city council

2. Climate Central Coastal Risk Screening Tool

- **Application**: Population exposure analysis and public communication
- **How it was used**: Generated compelling visuals showing familiar landmarks underwater. Used in community meetings, media briefings, and grant applications.
- **Outputs**: Maps showing 17,600 residents living in areas at risk from 3 feet of sea level rise; featured in local news coverage

3. NOAA Digital Coast Sea Level Rise and Coastal Flooding Impacts Viewer

- **Application**: Economic impact assessment
- **How it was used**: Estimated property values, tax revenue, and economic activity at risk. Analyzed impacts to roads, utilities, and public facilities.
- **Outputs**: $8.2 billion in property value at risk; 60+ miles of roads requiring elevation or protection

4. FEMA Flood Maps & Risk MAP

- **Application**: Current flood insurance and hazard baseline
- **How it was used**: Compared FEMA 100-year floodplains with sea level rise projections to identify areas where risk is increasing beyond current maps.

- **Outputs**: Identified 3,200 properties outside FEMA zones but at high future risk

5. SOVI (Social Vulnerability Index - CDC)

- **Application**: Environmental justice and equity analysis
- **How it was used**: Overlaid flood risk maps with socioeconomic vulnerability data to identify disadvantaged communities facing disproportionate climate impacts.
- **Outputs**: Found that predominantly Black neighborhoods in low-lying areas had highest vulnerability; informed equity-focused adaptation priorities

Planning & Design Tools

6. CoastalResilience.org (The Nature Conservancy)

- **Application**: Green infrastructure opportunity mapping
- **How it was used**: Modeled how wetland restoration, living shorelines, and tree planting could reduce flood exposure and provide co-benefits.
- **Outputs**: Identified 450 acres of potential wetland restoration; calculated $12M in avoided damages from nature-based solutions

7. i-Tree Tools (USDA Forest Service)

- **Application**: Urban forest stormwater benefits quantification
- **How it was used**: Assessed existing tree canopy and modeled expansion scenarios. Calculated stormwater interception from increased tree planting.
- **Outputs**: Urban forestry plan targeting 40% tree canopy cover; projected 58 million gallons/year additional stormwater capture

8. Adaptation Clearinghouse (Georgetown Climate Center)

- **Application**: Best practice research and peer city learning
- **How it was used**: Searched database for adaptation strategies from similar coastal cities. Connected with planners in Charleston, Miami Beach, and Annapolis.

- **Outputs**: Informed policy approaches including floodplain buyouts, resilient design standards, and stormwater fees

Community Engagement Tools

9. NOAA Climate Explorer

- **Application**: Local climate projection communication
- **How it was used**: Generated accessible graphs showing projected increases in extreme precipitation and hot days alongside sea level rise at community workshops.
- **Outputs**: Helped residents understand that flooding risk comes from multiple sources, not just sea level rise

10. Mapbox & ArcGIS Online (web mapping)

- **Application**: Interactive public-facing flood risk maps
- **How it was used**: Created "Norfolk FloodWise" web portal where residents could search addresses, view flood risk, and explore adaptation options.
- **Outputs**: 12,000+ unique visitors in first year; increased public awareness and preparedness

Implementation Process

Phase 1: Assessment (2014-2015)

The city started by using NOAA and Climate Central tools to establish baseline risk. Key activities:

1. **Internal capacity building**: Trained staff from multiple departments on climate tools
2. **Data gathering**: Downloaded elevation data, assembled flood history, compiled infrastructure inventories
3. **Initial modeling**: Ran scenarios for 1, 2, 3, and 5 feet of sea level rise
4. **Vulnerability assessment**: Created risk maps for buildings, roads, utilities, parks

5. **Economic analysis**: Calculated replacement costs and tax revenue impacts

Result: Comprehensive Recurrent Flooding Report documenting 5,600 acres at risk

Phase 2: Strategy Development (2015-2017)

Armed with data, Norfolk developed its comprehensive resilience strategy:

1. **Vision 2100**: Long-term adaptation framework looking 80 years ahead
2. **Ohio Creek Watershed Plan**: Detailed plan for most vulnerable neighborhood using CoastalResilience.org to model hybrid gray-green infrastructure
3. **Coastal Resilience Strategy**: Citywide approach balancing protection, accommodation, and retreat
4. **Zoning updates**: New floodplain regulations requiring higher elevations for new construction
5. **Capital improvements plan**: $2.7B in flood infrastructure investments prioritized using risk data

Result: Council-adopted resilience plan with specific projects, timelines, and funding sources

Phase 3: Implementation & Monitoring (2017-Present)

Norfolk began implementing priority projects with continued tool use:

Major Projects:

- **Ohio Creek Living Shoreline** ($112M): 4,000 feet of living shoreline with marsh restoration; NOAA tools showed 2-foot storm surge reduction
- **Hague Floodwall** ($88M): Deployable floodwall protecting 7,500 residents; designed to 2070 sea level projections from NOAA data
- **Pretty Lake Resilience Park** ($15M): Converted flood-prone park to wetland with stormwater storage; i-Tree modeling showed 8M gallons/year capacity
- **Chesterfield Heights Buyouts** ($22M): Voluntary acquisition of 123 repetitively flooded homes in environmental justice community

- **Smart scale stormwater sensors**: Real-time flood monitoring network with public dashboard

Ongoing Monitoring:

- Annual updates using NOAA sea level trends
- Quarterly flood occurrence tracking
- Biennial resilience plan reviews with updated projections
- Community resilience surveys

Results & Outcomes

Quantitative Impacts

Flood Risk Reduction:

- 17,000 residents protected by completed projects (as of 2024)
- 850 properties removed from future flood risk through buyouts and elevation
- $340M in avoided damages over 20 years (preliminary estimate)
- 42% reduction in tidal flooding in Ohio Creek neighborhood

Economic Benefits:

- $340M in federal grants secured using tool-generated data
- $180M in private investment leveraged in protected areas
- Maintained $8B tax base that was at risk
- 3,200 jobs created/retained through resilience construction

Environmental Co-benefits:

- 35 acres of wetlands restored
- 450 acres of new/protected green space
- 15,000 trees planted (on track to 40% canopy goal)
- Improved water quality in Elizabeth River

Qualitative Outcomes

Institutional Change:

- Created Chief Resilience Officer position
- Established inter-departmental resilience team
- Integrated climate projections into all capital planning
- Updated comprehensive plan with resilience elements

Community Awareness:

- 87% of residents aware of flooding risk (up from 34% in 2014)
- Active neighborhood flood preparedness groups in 12 communities
- "Resilience storytelling" program documenting resident experiences
- Annual Resilience Festival attracting 5,000+ attendees

National Recognition:

- Rockefeller 100 Resilient Cities member
- NOAA Climate Adaptation Leadership Award (2019)
- APA National Planning Excellence Award (2020)
- Featured in National Climate Assessment case studies

Equity Progress:

- 60% of resilience funding directed to historically underserved neighborhoods
- Community-led design process for Chesterfield Heights buyouts
- Multilingual flood warning system
- Resilience job training program for low-income residents

Challenges & Lessons Learned

Challenges Encountered

1. Data Limitations:

- Initial FEMA maps underestimated risk; had to educate stakeholders about differences between FEMA and sea level rise projections

- Hyperlocal drainage issues not captured by regional models
- **Solution**: Partnered with Old Dominion University for neighborhood-scale modeling; invested in LiDAR data collection

2. Funding Gaps:

- $2.7B need far exceeds city budget capacity
- Federal grant cycles don't align with urgent local needs
- **Solution**: Diversified funding (federal, state, private, bonds); phased implementation over 20 years

3. Community Displacement Concerns:

- Buyout programs raised fears of forced relocation and gentrification
- Historic Black neighborhoods concerned about loss of community
- **Solution**: Voluntary programs only; community-led planning; affordable housing commitments in protected areas

4. Technical Capacity:

- City staff initially lacked GIS and climate science expertise
- Consultant costs strained budgets
- **Solution**: Staff training; university partnerships; peer city knowledge exchange

5. Political Will:

- Early skepticism from some council members about climate investments
- Competing priorities (schools, public safety)
- **Solution**: Focused messaging on current flooding (not future climate); demonstrated ROI; secured quick wins

Key Lessons Learned

Start with Free Tools: Public tools like NOAA and Climate Central were essential for initial assessments without large budgets. They provided sufficient data to justify hiring consultants for detailed studies.

Visualizations Matter: Maps showing familiar places underwater were far more persuasive than statistics. Climate Central's comparison slider was particularly effective in community meetings.

Multiple Tools Needed: No single tool did everything. Norfolk used 10+ different platforms, each serving specific purposes. Integration was manual but valuable.

Data Drives Funding: Grant applications required quantitative risk data. NOAA and FEMA tools provided credible, peer-reviewed data that federal agencies accepted.

Community Engagement is Critical: Early resistance turned to support when residents saw risk data for their own neighborhoods and helped design solutions. The FloodWise web portal enabled self-service exploration.

Equity Must be Intentional: Without deliberate analysis using SOVI and community input, default engineering solutions would have prioritized downtown over vulnerable neighborhoods.

Monitor and Adapt: Sea level rise projections have increased since 2014. Annual updates using latest NOAA data ensure plans remain relevant.

Green + Gray Works: Hybrid approaches using CoastalResilience.org modeling proved more cost-effective and popular than seawalls alone.

Transferability to Other Cities

What's Replicable

✅ **Assessment Framework**: Other cities can follow Norfolk's phased approach: screen with free tools → detailed analysis → community engagement → implementation

✅ **Tool Combination**: The specific mix of tools (NOAA, Climate Central, CoastalResilience, i-Tree) works for any U.S. coastal city

✅ **Governance Structure**: Chief Resilience Officer + interdepartmental team model is scalable

✅ **Equity Analysis**: Overlaying SOVI with flood risk is applicable anywhere

✅ **Public Portal**: FloodWise concept can be replicated with Mapbox/ArcGIS Online

Context-Specific Elements

⚠️ **Military Presence**: Navy partnership is unique to Norfolk
⚠️ **Subsidence Rate**: Norfolk's land sinking is unusually rapid; other cities may have more/less time
⚠️ **Political Context**: Virginia's state climate policies supported local action; varies by state
⚠️ **Funding Access**: Proximity to Washington D.C. helped secure federal grants

Recommendations for Other Cities

Small Cities (<50k): Start with NOAA SLR Viewer and Climate Central. Focus on one vulnerable neighborhood. Seek state/regional support.

Medium Cities (50-250k): Follow Norfolk's phased approach. Apply for federal grants using free tool data. Partner with universities.

Large Cities (>250k): Can afford detailed consultants but use free tools for initial screening and community engagement. Share expertise with smaller neighbors.

Future Plans

Norfolk continues to evolve its resilience approach:

Near-Term (2025-2030):

- Complete Hague District floodwall system
- Expand living shoreline network by 8,000 linear feet
- Launch resilience bond program ($400M)
- Update resilience plan with latest NOAA projections
- Pilot floating architecture in flood zones

Long-Term (2030-2050):

- Managed retreat from lowest-lying neighborhoods
- Regional surge barrier feasibility study (with Army Corps)
- Climate-adapted affordable housing development
- 100% renewable energy for resilience infrastructure

- Climate migration preparedness (Norfolk may receive climate migrants from other areas)

Tool Needs:

- Better models for compound flooding (rain + tide + surge)
- Adaptation cost-benefit calculators
- Social cohesion impact assessment tools
- Migration pattern projection tools

Resources & References

Project Documents

- Norfolk Vision 2100 Plan - Full resilience strategy
- Ohio Creek Watershed Study - Detailed neighborhood plan
- FloodWise Portal - Public flood risk tool

Data Sources

- NOAA Tides & Currents: Sewells Point tide gauge data
- Climate Central: Population exposure analysis
- FEMA: Flood insurance claim history
- USGS: LiDAR elevation data

Academic Publications

- Kleinosky, L.R., et al. (2018). "Vulnerability assessment and adaptive capacity in Norfolk, Virginia." *Ocean & Coastal Management*, 161, 14-26.
- Mitchell, M., et al. (2017). "Recurrent flooding study for Tidewater Virginia." *Virginia Institute of Marine Science*.

Media Coverage

- Washington Post: "How Norfolk is adapting to rising seas" (2019)
- NPR: "A Virginia city's $2 billion bet on resilience" (2020)

- Scientific American: "Norfolk's experiment in managed retreat" (2022)

Awards & Recognition

- 100 Resilient Cities member (2015-2019)
- NOAA Climate Adaptation Leadership Award (2019)
- APA National Planning Excellence Award for Best Practice (2020)

Contact Information

City of Norfolk Resilience Office

- Chief Resilience Officer: [name]
- Email: resilience@norfolk.gov
- Website: www.norfolk.gov/resilience
- Twitter: @NorfolkVA

Case Study Authors

- [Your name], Repository Curator
- Date: January 2026

Tool Questions
For questions about specific tools mentioned:

- NOAA tools: coastal.questions@noaa.gov
- Climate Central: info@climatecentral.org
- CoastalResilience.org: coastalresilience@tnc.org

Case Study Type: Municipal Coastal Resilience Planning

Primary Tools: NOAA Sea Level Rise Viewer, Climate Central, CoastalResilience.org, i-Tree

Investment Level: $2.7 billion over 25 years

Replicability: High for coastal cities; moderate for inland flood contexts

Back to Case Studies | Back to Main